

DEBUUGING REPORT DOCUMENTATION

First of all I started to run the code by cmake on Ubuntu-bec I'm friendlier with because of my robotics projects. Then I faced the very first error that the project doesn't compiled because of missing heading `#include <object>` in `/include/safety_monitor.hpp` code

```
n file included from /home/sara/ARL-solomission/debugging/main.cpp:1:
/home/sara/ARL-solomission/debugging/include/safety_monitor.hpp:43:6: error: 'optional' in namespace 'std' does not name a template type
43 | std::optional<Obstacle> FindNearestObstacle(const std::vector<Obstacle>& obstacles);
    | ^
/home/sara/ARL-solomission/debugging/include/safety_monitor.hpp:5:1: note: 'std::optional' is defined in header '<optional>'; did you forget to '#include <optional>'?
4 | #include <vector>
+++ |+#include <optional>
5 |
/home/sara/ARL-solomission/debugging/main.cpp: In function 'int main()':
/home/sara/ARL-solomission/debugging/main.cpp:21:31: error: 'findNearestObstacle' is not a member of 'arl'
21 |     const auto nearest = arl::findNearestObstacle(obstacles);
    |                               ^
make[2]: *** [CMakeFiles/rover_safety_monitor.dir/build.make:76: CMakeFiles/rover_safety_monitor.dir/main.cpp.o] Error 1
make[1]: *** [CMakeFiles/Makefile2:83: CMakeFiles/rover_safety_monitor.dir/all] Error 2
make: *** [Makefile:91: all] Error 2
sara@sara-laptop: ~/debugging$ ./build/rover_safety_monitor
```

Also after that I checked the cmakefile to discover that the `/src` directory didn't get in the executables so I added it and the project compiled give me those outputs which were totally different that the outputs in the assignment.

```
[100%] Built target rover_safety_monitor
sara@sara-laptop: ~/debugging$ ./build/rover_safety_monitor
ARL Rover Safety Monitor
Valid detections: 1
Nearest obstacle: sign
World position: (79.4, 80.4)
Stopping distance: 165.6 m
Emergency brake: clear
```

So I started one by one why the detections are only 1?

First of all there was a huge mistake in the confidence test it was `0.0 < x < min config confidence` which was logically wrong and also wrong for the specifications of the project so I fixed it to `min < x < 1`

Then I got this output which means there's still an obstacle didn't get to be detected

```
sara@sara-laptop: ~/debugging$ ./build/rover_safety_monitor
ARL Rover Safety Monitor
Valid detections: 2
Nearest obstacle: crate
World position: (88.4, 72.6)
Stopping distance: 165.6 m
Emergency brake: clear
```

So I discovered that there's a logic of `index+1 < size(5 no of obstacles)` which make the last index to to be counted in the logic (`4+1 < 5` not true at all)so I removed the `+1` thing then I got finally the first correct output!

```
sara@sara-laptop: ~/debugging$ ./build/rover_safety_monitor
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: crate
World position: (88.4, 72.6)
Stopping distance: 165.6 m
Emergency brake: clear
```

So I moved forward to the second output the nearest obstacle→so I checked the distance comparison function and there was a wrong logic of compare make the rover not to consider the nearest obstacle as it takes 25 and compare it with the obstacles (12.6>25 for cone so not counted which totally wrong) so I changed the sign and the 2nd output is corrected too!

```
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: cone
World position: (98.2, 58.9)
Stopping distance: 165.6 m
Emergency brake: clear
```

Here we go for distance calculation issue that gives wrong world coordinates→the huge mistake lies in the angles that get into the sin&cos weren't converted to radian specially they are taken as degrees there wasn't any function to convert it was taken as rad=degree ! so I fixed it by adding the function and the y coordinate have got corrected!

```
sara@sara-laptop:~/debugging$ ./build/rover_safety_monitor
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: cone
World position: (104.0, 62.0)
Stopping distance: 165.6 m
Emergency brake: clear
```

For x coordinate the problem was in that the left was added not subtracted so I fixed it and finally the 3rd output is now correct!

```
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: cone
World position: (96.0, 62.0)
Stopping distance: 165.6 m
Emergency brake: clear
```

So coming to stopping distance → the velocity was taken the same for km/h like m/s=km/hr ,Which of course is wrong and makes a huge error so I fixed the function by converting the m/s to km/hr and the 4th output is now with us!

```
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: cone
World position: (96.0, 62.0)
Stopping distance: 20.0 m
Emergency brake: clear
```

The last output was wrong because there wasn't any logic for emergency braking so it will always give clear ! I fixed it by adding a function and finally!!! **All the outputs is Correct!!!**

```
ARL Rover Safety Monitor
Valid detections: 3
Nearest obstacle: cone
World position: (96.0, 62.0)
Stopping distance: 20.0 m
Emergency brake: ENGAGE
```

There were totally 2 syntax errors and 7 logic errors as a total result all fixed and it was fun to debugging them!!

For Ai Usage I used it to understand the syntax because I'm more comfort with python and not advanced on C++ (I'm working on it this period) the Ai tool is chat gpt.